ISAAI - Automatic Report Documentation

Information Systems Architecture, SoSe 2026

Final Report

Module: ISA (Information Systems Architecture)
University: Frankfurt University of Applied Sciences — Faculty 2
Semester: Summer Semester 2026
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Table of Contents

1. Introduction

The **ISAAI (Information Systems Architecture and Integration)** project was conducted as part of the modules *ISA (Information Systems Architecture)* and *A+I (Architecture and Integration)* at the Frankfurt University of Applied Sciences in the summer semester of 2026. It addresses the automation of the daily processing of structured financial reports received as XML files in ZIP archives via E-Mail.

1.1 Project Context

In the daily business operations of a financial service provider, two types of reports (Report Type A — Daily Trades and Report Type B — Monthly Trader Summaries) are received daily from a trading platform. These reports contain compliance-relevant trading and trader data, which must be checked for rule violations and subsequently archived.

The manual process (extraction, compliance check, evidence archiving and supervisor escalation) is replaced by a automated Microsoft Power Automate Flow.

1.2 Dual Module Approach

The project pursues a dual-module approach that integrates the requirements of both courses:

Module	Focus	Technology
A+I	Enterprise Architecture, ArchiMate Modeling, Invoice, Chart Automation	ArchiMate, Power Automate, Powerpoint
ISA	Operative Process Automation, XML-Validation, Exception Governance, GOAL BPMN	Power Automate, Powerpoint, Camunda Modeler

1.3 Project Goal (SMART)

Specific	Measurable	Achievable	Relevant	Time-bound
Report A/B XML Pipeline + parallel ISA Invoice Pipeline	End-to-End Automation for the standard path; Exceptions measurable	GitHub CI/CD, Mock APIs, existing BPMN/ArchiMate models	White-box interfaces; Audit Trail; 30–40 min/day time savings	Deadline June 30, 2026

1.4 Project Organization

Role	**Person / Unit**
Sponsor and Academic Lead	Dominik Dietrich — FRA UAS / FB2
Architecture and Software Engineering	Christina Malki, Altay Hennig — Modules ISA and A+I
Operative Owner	Sarah Bullinger — Project Management Operations

2. Management Summary

2.1 Initial Situation

The existing process for the daily processing of financial reports (Report Type A and Report Type B) requires **30–40 minutes of manual effort** every day. An employee opens incoming emails, extracts ZIP attachments, imports XML data into a spreadsheet, visually checks thresholds, and manually creates evidence documents. This process carries significant risks: incorrect field selection, inconsistent results, forgotten archiving steps, and a lack of system-enforced upload-to-sign-off processes.

2.2 Solution

The ISAAI system replaces the entire manual process with a fully automated **Microsoft Power Automate Cloud Flow**, which covers the following steps:

1. **E-Mail Trigger:** Automatic detection of incoming emails with the subject "Financial Report" and a ZIP attachment.
2. **ZIP Extraction and XML Parsing:** Automatic unpacking and parallel validation of both report types.
3. **Rule-Based Compliance Check:** XPath-based evaluation of violation markers (DayViol and MonthViol).
4. **Evidence Generation:** Automatic creation of professional XLSX evidence documents.
5. **Exception Governance:** Supervisor escalation only when violations are detected.
6. **SharePoint Audit Trail:** Seamless documentation of all processing steps.

2.3 Result

Full automation eliminates daily manual effort while ensuring full auditability and human control for exceptions.

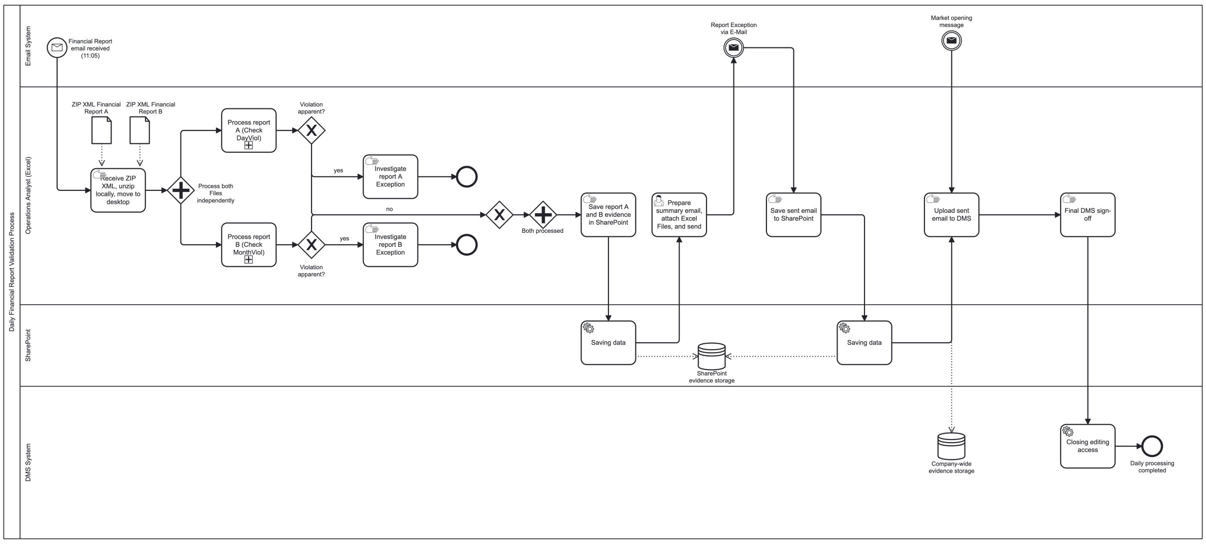
3. Problem Description

3.1 As-Is State (CURRENT State)

In the current state, daily financial reports are processed entirely manually. The process includes the following steps:

1. **E-Mail Receipt:** An employee receives daily E-Mails with ZIP attachments containing XML reports.
2. **Manual Extraction:** ZIP archives are manually unpacked and the contained XML files are reviewed.
3. **Table Import:** The XML data is manually imported into a spreadsheet.
4. **Visual Check:** Thresholds and violation markers are checked visually.
5. **Evidence Creation:** Evidence documents are manually created and formatted.
6. **Archiving:** Documents are filed and approved in the document management system.

BPMN CURRENT — Manual Daily Flow



3.2 Identified Weaknesses

Weakness	**Description**	**Impact**
High Time Expenditure	30–40 minutes of daily manual effort	Loss of productivity, resource binding
Media Break	Switching between Collaboration Storage and Document Repository	Susceptibility to errors, loss of information

Lack of System Enforcement	No system-supported upload-to-sign-off process	Compliance risk, lack of traceability
Human Errors	Incorrect fields, inconsistent outputs, forgotten steps	Quality defects, audit risks
Scaling Problem	Manual process does not scale with increasing report volume	Capacity bottlenecks during peak loads

3.3 Risk and Control Gap

Manual processing multiplies traceability gaps and increases the risk of unauthorized changes or overlooked violations. The lack of automation and a non-standardized audit trail exacerbate this problem.

3.4 Report Types

Two report types are processed daily, each with a specific compliance marker:

Attribute	**Report Type A (Daily)**	**Report Type B (Monthly)**
Coverage Period	Previous business day (Trade-Level Detail)	Month-to-Date cumulative trader summary
Delivery Frequency	Daily, at the agreed processing time	Daily, together with Report Type A
Violation Marker	DayViol	MonthViol
Purpose	Detection of same-day trade exceptions	Tracking monthly cumulative trader breach trends

4. Solution Design

4.1 Architecture Evolution: CURRENT → GOAL

The project utilizes already realized automation capabilities that are provided by Microsoft alongside their office suite. The manual labor can be systematically replaced as the transformation from the manual as-is state to the automated target state takes place.

4.2 GOAL (Realized State)

The GOAL state represents the actually implemented solution — the full Power Automate Flow with SharePoint integration, parallel XML validation, and automatic XLSX evidence file generation. The DMS integration was deliberately removed from the scope; SharePoint serves as the sole Evidence Repository.

4.2 Power Automate Flow — Technical Architecture

The flow **ISAAI – Daily Report Processing** is structured into six phases and is available as an importable Power Automate package in the repository under:

src/flow/ISAAI-DailyReportProcessing_20260625165442.zip

4.5 Technology Stack

Component	Technology
Automation	Microsoft Power Automate (Cloud Flow)
Data Storage	SharePoint Online (Lists + Document Library)
Evidence Format	XLSX (Excel)
Architecture Modeling	ArchiMate 3.1 (Archi)
Process Modeling	BPMN 2.0
Local Simulation	Python 3, openpyxl, python-pptx
Version Control	Git / GitHub
Artificial Intelligence	Gemini, Claude, Antigravity, CursorAI

5. Challenges

5.1 Technical Challenges

Power Automate

The Microsoft Copilot import could not generate nor import complex branching flows flawlessly, so the flow had to be built manually in the Power Automate Designer (see docs/guides/SHAREPOINT_FLOW_GUIDE.md).

5.2 Architectural Challenges

DMS Integration (Out of Scope)

The original VISION envisioned integration with the company's internal Document Management System (DMS), including UI scraping/RPA. This integration was deliberately removed from the GOAL scope because:

- UI scraping is fragile and requires high maintenance
- No stable API to the DMS was available
- SharePoint fully meets the requirements as the sole Evidence Repository

6. Lessons Learned and Findings

6.1 Architecture and Modeling

ArchiMate usage may differ between internal strategy and delivery. The ArchiMate diagrams have proven to be an excellent communication tool to transparently present the transformation path from the manual to the automated solution. Compartmentalization in internal communication can increase efficiency and understanding.

6.2 Technology and Automation

Power Automate is suitable for standardized processes. For the defined use case — E-Mail trigger, ZIP extraction, XML parsing, rule-based validation, and SharePoint storage — Power Automate has proven to be a suitable low-code platform.

7. Outcome and Conclusion

7.1 Project Result

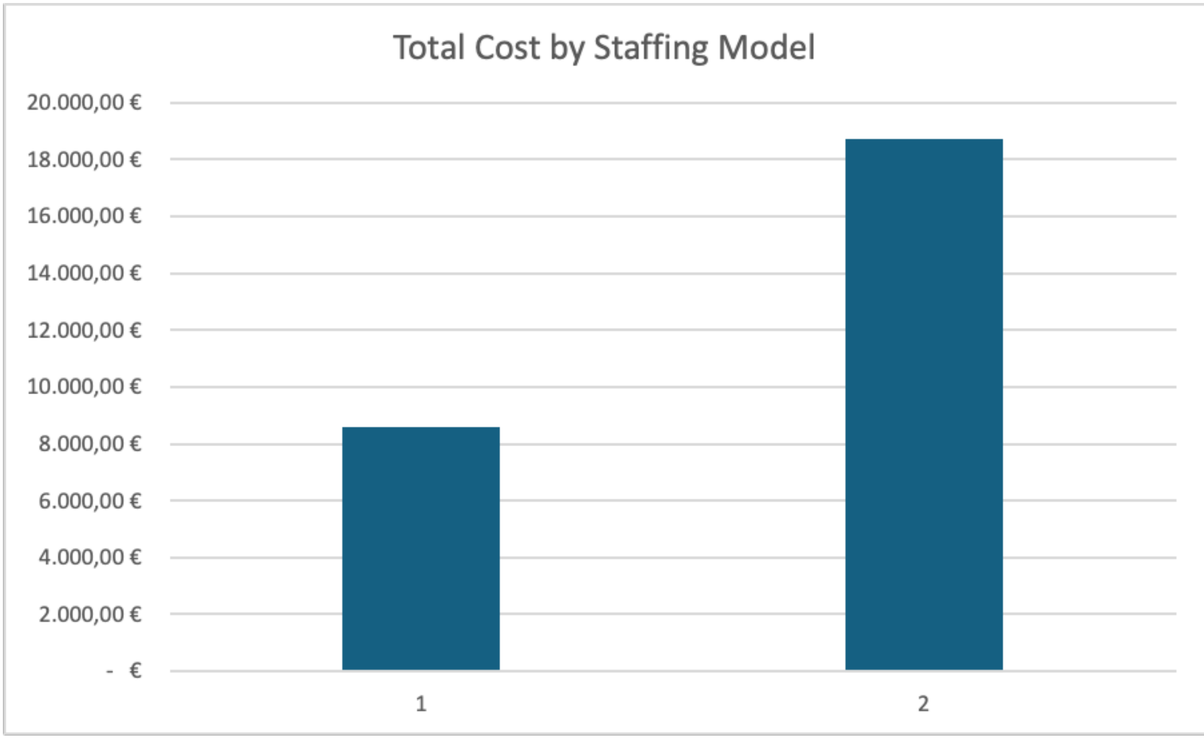
The ISAAI project achieved all defined goals:

Goal	Status	Description
Full automation of the standard path	Achieved	E-Mail, ZIP, XML, Validation, Evidence, SharePoint, Notification
Exception Governance	Achieved	Human-in-the-loop for violations with supervisor escalation

Audit Trail	Achieved	Seamless documentation in SharePoint list + document library
ArchiMate Modeling (3 stages)	Achieved	CURRENT, VISION, GOAL each as ArchiMate models (.archimate)
BPMN Process Modeling (3 stages)	Achieved	CURRENT, VISION, GOAL each as BPMN diagrams (.bpmn)
Power Automate Flow	Achieved	Importable flow package in the repository
Local Simulation	Achieved	30 reproducible test runs with Python scripts
ISA Module (parallel)	Achieved	Invoice receipt, JSON/CSV, Chart Automation
Presentation Material	Achieved	Consolidated Findings Deck + Module Presentations

7.2 Quantitative Benefits

Category	**Recommended Percentage**	**Estimated Cost (Student Team)**	**Estimated Cost (Regular Developer)**	**Notes**
Developer Cost		**7.428**	**16.147,90**	
Project Management & Coordination	**15%**	1114,2	2422,185	These are rough estimates
Change Management & Training	**5%**	55,71	121,10925	These are rough estimates
Contingency Buffer	**10%**	5,571	12,110925	These are rough estimates
SUM		**8.603,48 €**	**18.703,31 €**	



The cost calculation provides an initial estimate for automating selected steps of the report evaluation process related to the assessment of a violation. Based on the Function Point and COCOMO methods, the implementation effort is estimated at approximately 323 working hours.

The figure 8.603,48€ represents an initial planning estimate and includes additional allowances for project coordination, change management, training, and contingency.

7.4 Conclusion

The ISAAI project successfully demonstrates how the combination of enterprise architecture methodology (ArchiMate, BPMN) and low-code automation (Power Automate, SharePoint) can completely transform an operational process. The systematic documentation of the transformation path CURRENT, VISION, GOAL ensures that architectural decisions remain traceable and the solution remains maintainable.

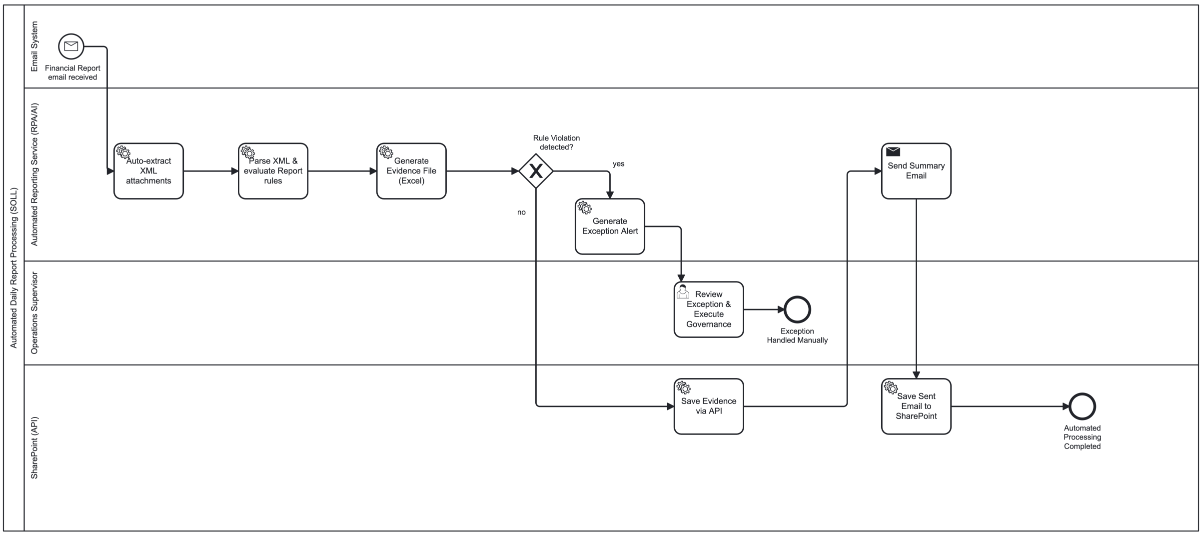
The deliberate scope reduction from VISION to GOAL (excluding DMS integration) also shows that pragmatic architecture decisions are just as important in a university project as they are in corporate practice. The SharePoint-based solution meets all functional requirements while providing a solid foundation for future enhancements.

8. BPMN Model with Camunda

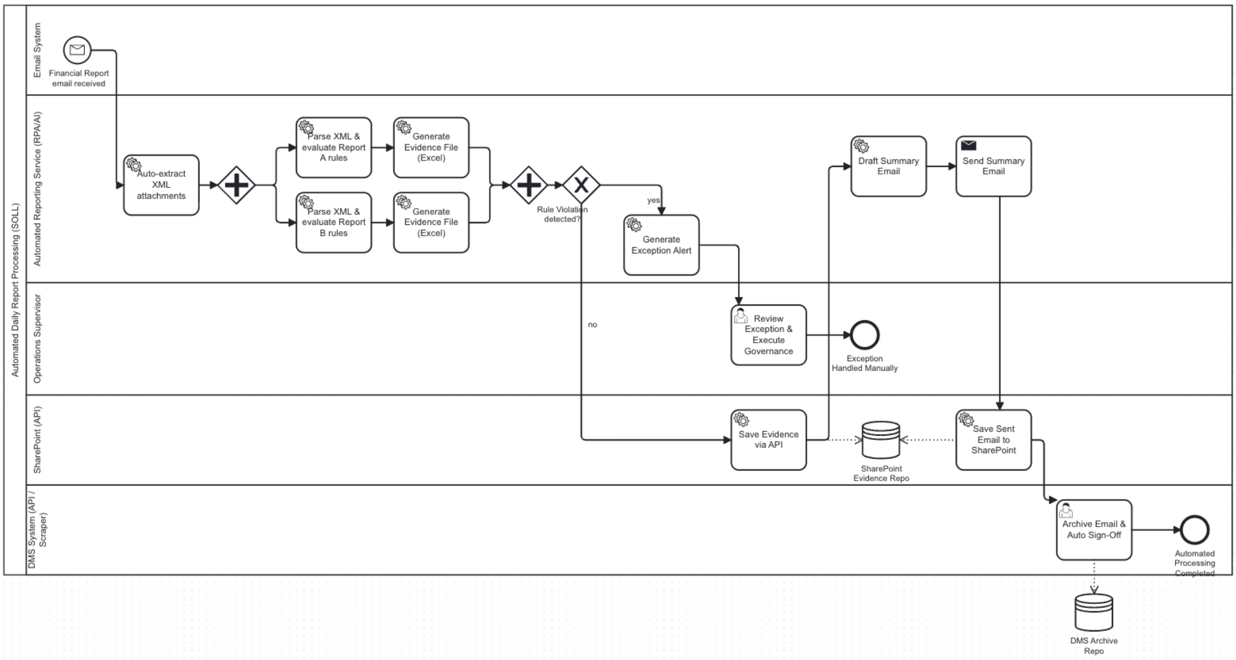
The project includes BPMN 2.0 process models for the three architecture stages. The models were created using Camunda and are available as .bpmn files in the repository.

The GOAL BPMN model describes the actually implemented process using Power Automate and SharePoint.

BPMN GOAL — Strategic Validation Process:



BPMN GOAL — Operational Validation Process:



9. Magic Charts

9.1 Concept

The **produced Evidence** is stored in processible formats and through a scripting module that automates the creation of visualizations and presentations from the processed report data. It transforms raw data from XML reports into professional, meaningful charts and presentation slides.

9.2 Implemented Scripts

1. **Test Data Generation (simulate_runs.py):** Generates reproducible mock XML/ZIP test data for 30 days:
2. **Local Flow Processing (local_flow_processor.py):** Simulates the full Power Automate Flow offline and generates professionally styled evidence files:
3. **Consolidated Findings Presentation (generate_findings_presentation.py):** Generates a consolidated findings presentation across all test runs:
4. **Boardroom Deck (build_boardroom_deck.py):** Creates a management-ready boardroom presentation:
5. **Demo Flow (demo_flow.py):** Interactive flow run with visual output:

10. Attachments

10.1 Central Project Documents

https://github.com/pollo60/-ISAAI-Automatic-Report-Documentation.git

Document	Path	Description
Project Charter (PDF)	docs/Documentation/Project_Charter_ISAAI.pdf	Formal project mandate
Project Charter (LaTeX)	docs/Documentation/Project-Charter-ISAAI.tex	Source code of the project mandate
Roadmap	docs/Documentation/roadmap.md	Phase plan with 5 phases
Issue List	docs/Documentation/issue-list.md	Task list with 5 milestones
SharePoint Flow Guide	docs/guides/SHAREPOINT_FLOW_GUIDE.md	Step-by-step setup
Flow Package	src/flow/ISAAI-DailyReportProcessing_*.zip	Importable Power Automate Flow
ISA Presentation	presentations/Project Presentation/praesi isa 2906.pptx.pptx	ISA module presentation
A+I Presentation	presentations/Project Presentation/praesi ai 2906.pptx.pptx	A+I module presentation

11. Sources

11.2 Architecture and Modeling

6. **ArchiMate 3.1 Specification.** The Open Group. https://pubs.opengroup.org/architecture/archimate3-doc/
7. **Archi — ArchiMate Modelling Tool.** https://www.archimatetool.com/
8. **BPMN 2.0 Specification.** Object Management Group (OMG). https://www.omg.org/spec/BPMN/2.0/
9. **Camunda BPMN Modeler.** Camunda Services GmbH. https://camunda.com/

11.3 Compliance and Governance

11. **EU AI Act — Regulation (EU) 2024/1689.** European Parliament and Council. https://eur-lex.europa.eu/eli/reg/2024/1689/oj
12. **General Data Protection Regulation (GDPR) — Regulation (EU) 2016/679.** European Parliament and Council. https://eur-lex.europa.eu/eli/reg/2016/679/oj

11.4 Development Tools

14. **Python 3 Documentation.** Python Software Foundation. https://docs.python.org/3/
15. **openpyxl — A Python library to read/write Excel 2010 xlsx/xlsm/xltx/xltm files.** https://openpyxl.readthedocs.io/
16. **python-pptx — Python library for creating PowerPoint files.** https://python-pptx.readthedocs.io/
17. **GitHub — Collaborative Development Platform.** https://github.com/

11.5 Project Resources

19. **ISAAI GitHub Repository.** https://github.com/pollo60/-ISAAI-Automatic-Report-Documentation
20. **SharePoint Site — ISAAI Daily Report Processing.** https://studfrauasde.sharepoint.com/sites/ISAAIDailyReportProcessing